Wafer Map Anomaly Detection on WM-811K

From Reconstruction Baselines to Pretrained Local Feature Modelling

50.039 Deep Learning · Y2026 · Group 08

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The Problem

Semiconductor fabs generate hundreds of thousands of wafer maps. Most are normal. A few are not.

- **Defects are rare and costly:** each bad wafer drives up yield loss
- **Defect types are unpredictable:** 9 known patterns, but new ones can emerge
- **Labelled defect data is scarce:** 3.1% of the dataset is labelled defective; the rest is unlabelled
- **Classification does not scale:** you cannot enumerate every failure mode at training time

> **Goal:** Learn what a normal wafer looks like. Flag anything that deviates.

The Dataset: WM-811K

811,457 wafer maps from the LSWMD dataset

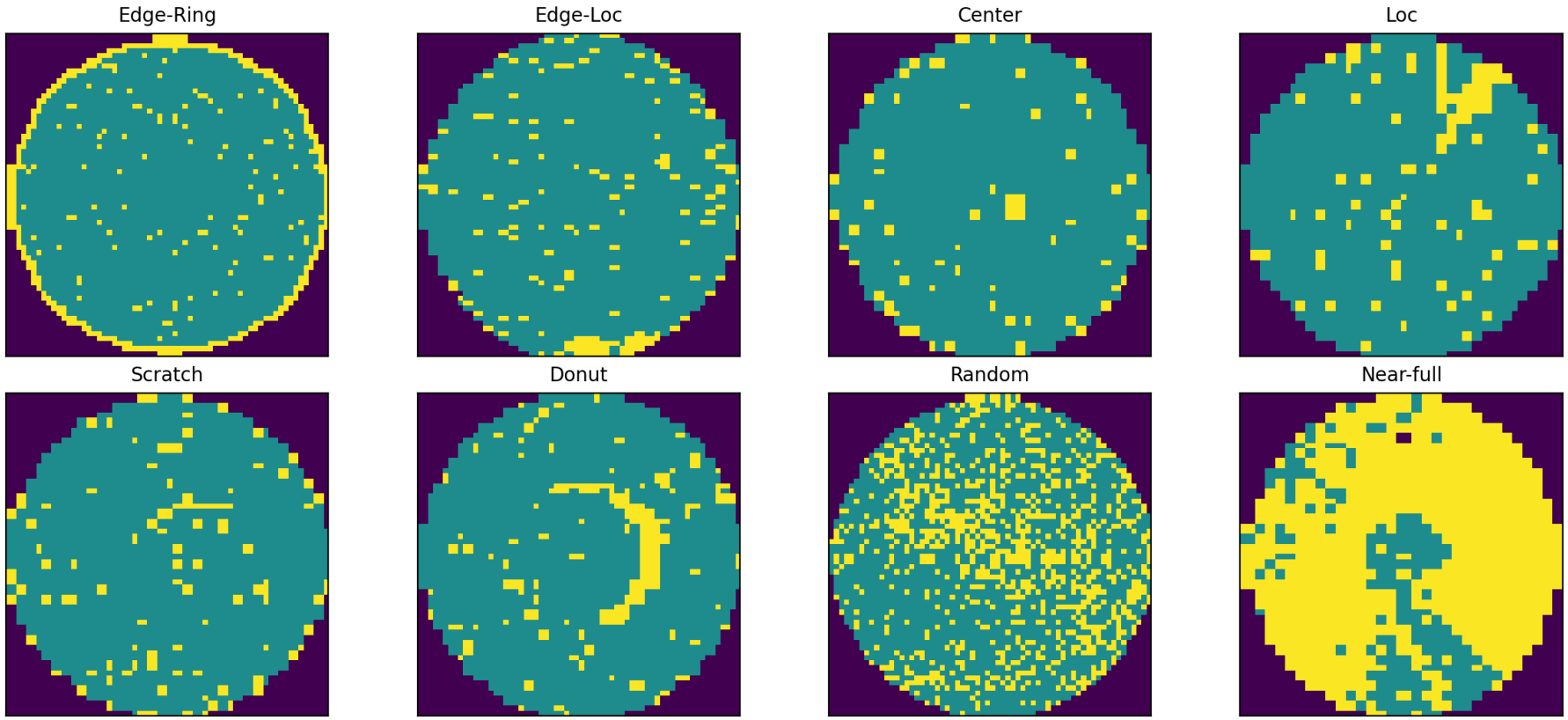
Split	Count	Share
Normal (labelled)	147,431	18.2%
Defective (labelled)	25,118	3.1%
Unlabelled	638,908	78.7%

Key challenge: Class imbalance: a model predicting every wafer as normal achieves 96%+ accuracy while catching zero defects.

The 9 defect pattern types differ in shape, size, and sparsity, making detection non-trivial.

Defect Type Gallery

Representative Defect Patterns from WM-811K Test Set



Our Plan: Normal-Only Training

We train **exclusively on normal wafer maps**. At inference, each wafer receives an anomaly score (higher means more deviant from normal).

Why not supervised classification?

- Requires balanced, labelled defect data: scarce in production
- Cannot detect defect types unseen at training time
- Anomaly detection generalises to new defect patterns automatically

Training plan:

- **Train:** normal wafers only
- **Validation:** normal wafers only (threshold calibration)
- **Test:** normal + all labelled defect types (held out completely)

How We Evaluate

Three metrics, because **accuracy alone is useless** for imbalanced data.

Metric	What it measures	Why it matters here
AUROC	Ranking quality across all thresholds	Threshold-free; strong ranking signal
AUPRC	Precision-recall area under curve	**Our primary metric** -- most informative at 5% prevalence
F1	Harmonic mean at the deployed threshold	Reflects real production performance

A model predicting all wafers as normal achieves **96%+ accuracy** and **zero defect recall**. AUROC and AUPRC expose this immediately.

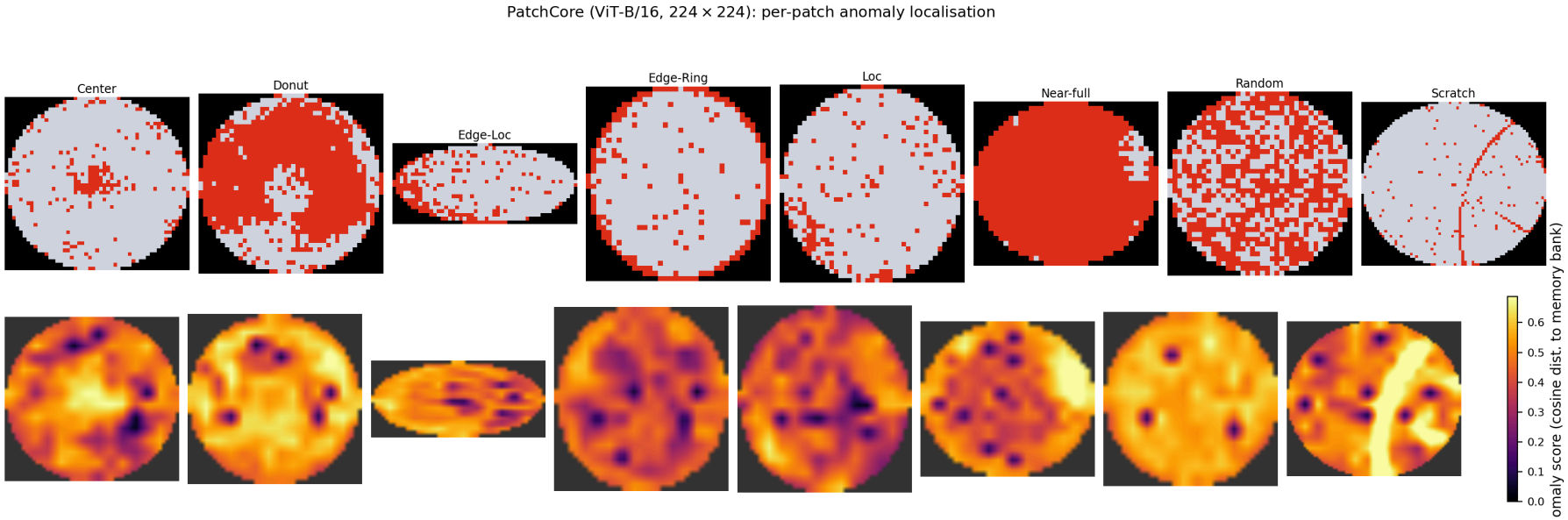
Threshold: 95th percentile of normal-only validation scores, no defect labels required.

Method Journey

Each method family was selected to address the limitations of its predecessor.

	Family	Hypothesis	Finding
H1	Reconstruction	Can pixel reconstruction detect anomalies?	Pixel errors too diffuse; sparse defects invisible
H2	Global Features	Do pretrained global features do better?	Pooling collapses spatial structure
H3	**PatchCore + CNN**	**Do local patch features matter?**	**Large jump: patch-level scoring is the key**
H4	PatchCore + ViT	Does backbone architecture matter?	ViT-B/16 at 224×224 wins
H5	PatchCore + ViT (MAE)	Does domain adaptation improve features?	AUPRC improves; F1 stays stable
H6	**Ensemble**	**Can diverse pretraining push further?**	**Best overall: AUROC 0.967**

PatchCore: Local Patch Scoring in Action



Top row: original wafer binary maps (red = failed dies). Bottom row: per-patch anomaly score heatmap at the ViT-B/16 patch grid resolution. Patch-level scoring localises small defects that pixel-averaged methods miss entirely.

Top 10 Models

64 training runs: Reconstruction — AE (11) · VAE (9) · SVDD (1) · Center-distance (2) · Teacher-Student (5) · FastFlow (3) · RD4AD (1) · PatchCore — CNN (14) · ViT (8) · DINOv2 (3) · Ensemble (5) · Supervised (1)

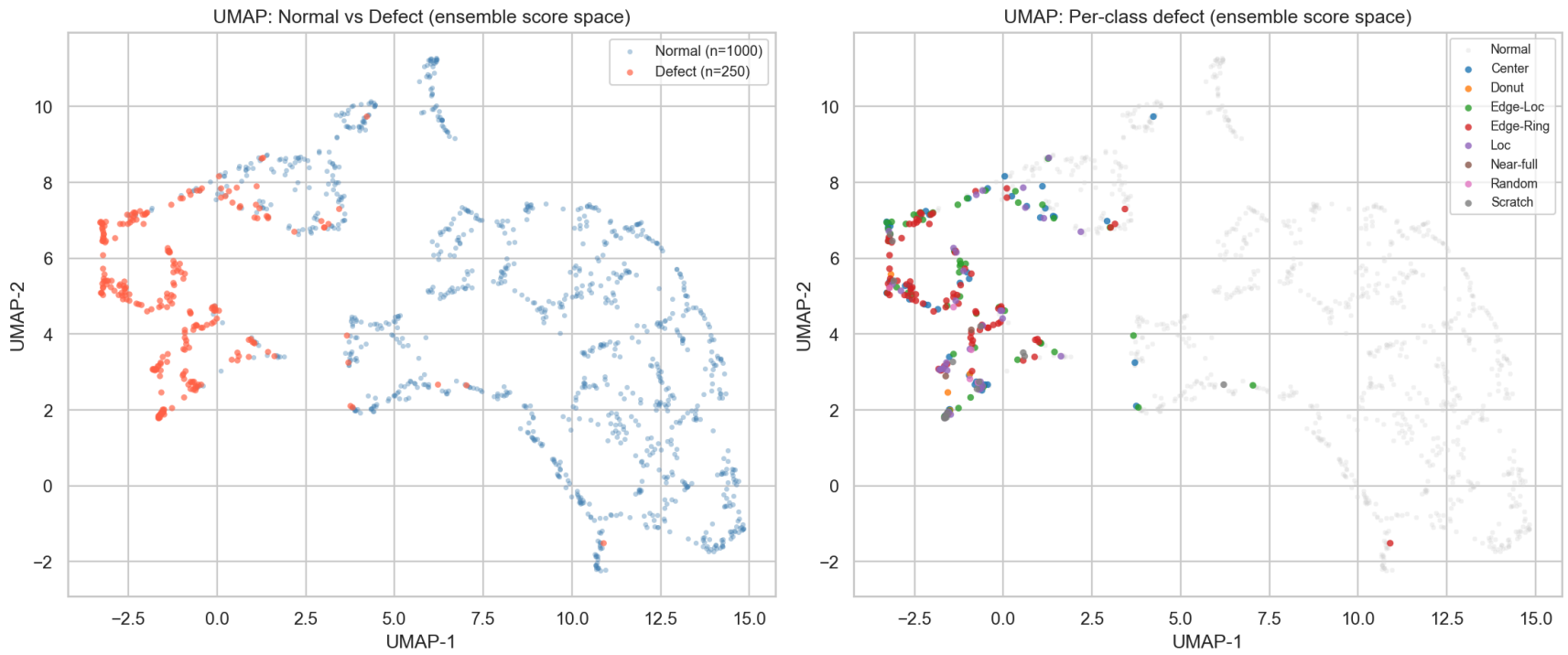
Rank	Family	Model	Sweep F1	AUROC	AUPRC
1	Ensemble	**ViT-B/16 + DINOv2 (max-fusion)**	**0.676**	**0.967**	0.716
2	PatchCore + ViT	ViT-B/16, MAE fine-tuned (25% mask)	0.692	0.962	**0.762**
3	PatchCore + ViT	ViT-B/16, MAE fine-tuned (75% mask)	0.668	0.959	0.717
4	PatchCore + ViT	ViT-B/16, frozen, 224×224	0.651	0.956	0.671
5	PatchCore + CNN	EfficientNet-B1, 240×240	0.653	0.935	0.609
6	PatchCore + CNN	WideResNet50-2, 224×224	0.634	0.931	0.659
7	Teacher-Student	ResNet50	0.606	0.909	0.599
8	Teacher-Student	WideResNet50-2	0.561	0.923	0.546
9	PatchCore + ViT	DINOv2 ViT-B/14, block-9	0.570	0.915	0.561
10	Reconstruction	Autoencoder, 224×224	0.587	0.901	0.596

ViT-B/16 vs DINOv2: Two Different Views of the Same Image

	ViT-B/16	DINOv2 ViT-B/14
Pretraining	Supervised ImageNet classification	Self-supervised self-distillation (no labels)
Objective	Predict the correct object class	Make semantically similar images cluster together
What it learns	Discriminative features that separate categories	Semantic features that capture visual similarity
Sensitivity	Structured, boundary-anchored defects	Texture and sparse local pattern deviations
Patch size	16×16 px (196 tokens per image)	14×14 px (256 tokens, finer granularity)
Scratch recall	0.67 (weaker on thin linear defects)	**0.93** (stronger on sparse, low-coverage patterns)
Loc recall	**0.79**	0.71

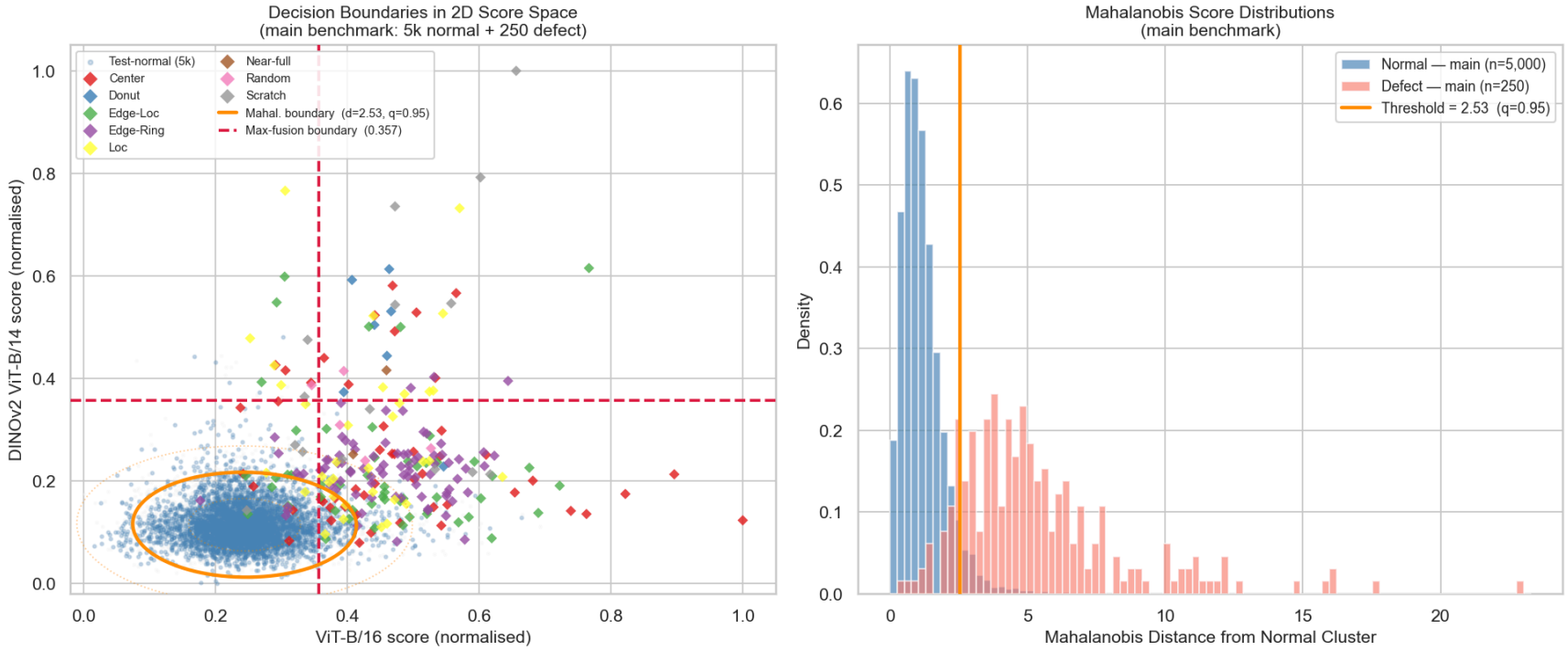
Because the two models fail on different defect types, taking the maximum anomaly score per wafer recovers detections that either model alone would miss.

Complementary Signals in Score Space



UMAP of [ViT score, DINOv2 score] per test wafer. Left: normal vs defect separation. Right: per-class clusters. Scratch concentrates where DINOv2 is high; Edge-Ring and Edge-Loc concentrate where ViT is high. Max-fusion captures both regions.

Mahalanobis Fusion: Tighter Decision Boundary in Score Space



Left: Mahalanobis ellipse (orange) vs max-fusion L-shape (red dashed) in normalised score space. Mahalanobis requires joint signal from both models. Right: score distributions show clear separation. Mahalanobis achieves AUROC 0.968, AUPRC 0.762, Sweep F1 0.722 — all above max-fusion.

Best Model: ViT-B/16 + DINOv2 Ensemble

Per-defect recall

Defect	ViT-B/16	DINOv2	Ensemble
Scratch	0.67	**0.93**	**0.80**
Loc	**0.79**	0.71	**0.91**
Edge-Loc	**0.79**	0.55	**0.85**
Center	**0.82**	0.56	**0.88**
Edge-Ring	**0.89**	0.74	**0.89**
Donut	1.00	1.00	**1.00**

Where one model is weak, the other is strong.

Overall metrics

Model	AUROC	AUPRC	F1
ViT-B/16	0.956	0.671	0.595
DINOv2	0.915	0.561	0.492
Ensemble (max-fusion)	0.967	0.716	**0.623**
Ensemble (Mahalanobis)	**0.968**	**0.762**	0.592

> Mahalanobis requires joint signal from both models. Better ranking (AUROC, AUPRC) but stricter threshold lowers deployed F1.

What If We Had Used Defect Labels?

Our entire approach assumes **normal-only training**. But what if that assumption costs us performance?

Two experiments to test this:

- **Defect-tuning PatchCore:** Fine-tune ViT-B/16 backbone layers with contrastive loss on defect examples
- **Supervised linear probe:** Train a classifier on up to 1,016 labelled defect samples across all defect classes — with **Scratch and Loc withheld** to test generalisation

> **The question:** Does knowing what defects look like actually help?

Result: Normal-Only Is the More Robust Choice

Supervised linear probe sweep (Scratch + Loc withheld from training):

Defect %	Train defects	AUROC	F1	Sweep F1	Scratch recall	Loc recall
0.1%	20	0.895	0.499	0.640	0.083	0.400
1.0%	203	0.941	0.554	0.721	0.333	0.657
5.0%	1,016	0.956	0.581	0.760	0.167	0.686
PatchCore (no labels)	**0**	**0.956**	**0.595**	**0.651**	**0.727**	**0.829**

With 1,016 labels, supervised AUROC matches PatchCore. But Scratch recall is **0.167** vs **0.727**: the classifier never saw Scratch during training. More labels only help for defect types you already have.

> **Conclusion:** Anomaly detection has no blind spots. It flags anything that looks unusual, whether or not that defect type existed at training time.